Answers and talk notes

Divide Decimals Using an Algorithm

What your student is learning

Tonight we divide when the DIVISOR is a decimal — like 9.6 ÷ 0.4 — by first turning it into an easier problem with the same answer.

Answers

Ask your student to explain before you look.

1. **A.** 10. Each place the point moves right makes the number 10 times as big, so 10.
2. **B.** 2 places to the right, to make 25. Slide 2 places right: 0.25 becomes 25, a whole number. So the answer is: 2 places to the right, to make 25.
3. **C.** 24 ÷ 6. Slide both points 1 place right: 2.4 ÷ 0.6 = 24 ÷ 6.
4. **D.** 7. 3.5 ÷ 0.5 = 35 ÷ 5 = 7.
5. **B.** 30. Slide both points 2 places right: 4.5 ÷ 0.15 = 450 ÷ 15 = 30.
6. **C.** 24. 7.2 ÷ 0.3 = 72 ÷ 3 = 24 cups.

Talk about it

Why do both decimal points slide the same number of places?

Home mission

Pretend a jug holds 9.6 liters and each cup holds 0.4 liter. Before dividing, ask: will there be more cups or fewer cups than 9.6? Why?

When they are stuck, ask

- Which number is the divisor? Is it whole yet?
- How many places did you move the divisor? Did the dividend move the same?
- We are dividing by less than one — should the answer be bigger or smaller?

Try not to say

- Move the decimal in the divisor and leave the other number alone.
- Dividing always makes the answer smaller.
- Just move one place, that is usually enough.